

USER MANUAL

BluetoothMax

Bluetooth LE gateway for ChessLink (Mode B) chess computer modules



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github.com/parlue/BluetoothMax

Contents

1. Overview
2. Supported e-boards
3. ManyaCynus special cases
4. Startup sequence
5. Standalone mode (no cable computer required)
6. Supported chess computer modules
7. Automatic PGN recording
8. Retrieving saved games
9. Firmware updates (web installer)
10. Hardware: components, wiring & safety
11. Version history
12. Contact, license & trademark notice

1. Overview

BluetoothMax is an independent Bluetooth LE gateway for chess computer modules that speak the ChessLink (Mode B) protocol over a cable. It replaces the cable between the chess computer module and a Bluetooth-enabled e-board with a Bluetooth LE connection — and, unlike a single-board bridge, it scans for and connects to **whichever supported e-board is present** at runtime, translating each board's own native protocol to and from Mode B on the fly. One firmware image, multiple board brands.

No cable chess computer at hand? The gateway can also run standalone, masquerading as a ChessLink or Chessnut board over BLE so wireless chess software can connect to it directly (see section 5).

```
Generic chess computer module with DIN connector
  ↕
  4-pin Mini-DIN
  ↕ RS-232
  HW-027 with MAX3232
  ↕ UART / 3.3 V
  ESP32-C3 SuperMini
  ↕ Bluetooth LE
  MILLENNIUM Supreme T2 BT -- or -- Chessnut Air/G0/Pro -- or -- ManyaCynus robot
```



2. Supported e-boards

Board	Status
MILLENNIUM Supreme T2 BT (and other genuine ChessLink boards)	Working — native Mode B relayed as-is
Chessnut Air / GO / Pro	Working — Chessnut's own BLE protocol translated to/from Mode B, including LED move suggestions and New Game/reset highlighting
ManyaCynus (camera-vision chess robot)	Working — ManyaCynus's own line-based BLE protocol translated to/from Mode B; decodes the chess computer's own LED move suggestions and commands ManyaCynus's arm to execute them, including castling, en passant and pawn promotion
iChessOne	Untested on real hardware — driver built from the vendor's own documented BLE protocol (Nordic UART Service), not yet confirmed against a real unit

Board detection is automatic: on every (re)connect attempt, the gateway scans for any known board's advertised BLE name and connects to whichever one it finds, with no manual board selection needed.

3. ManyaCynus special cases

ManyaCynus is a camera-vision robot, not a per-square sensor board, so its driver reproduces several behaviors of a real ChessLink board rather than having them natively.

Startup

The scanned position is checked against the normal and mirrored starting position before anything else; an incorrect setup is rescanned automatically every 5 seconds until it matches.

Options via the black King

With the board in its starting position, moving only the black King onto one of the following squares toggles a setting instead of being treated as a move — move it back to its home square (or the option's own "off" square) to leave the option again:

Black King square	Function
e5 / e6	Sound off / on
h5 / h6	Board-orientation flip on / off
d5 / d6	Free Analysis mode on / off
c5 / c6	Set Position mode on / off

Free Analysis mode

While active, every scanned position is forwarded to the chess computer as-is, without checking it's a legal move — the board is re-scanned automatically every 5 seconds. The connected chess software also needs its own support for this mode.

Castling, en passant and promotion

Reported to the chess computer as the individual piece lift/place steps a physical board would produce (e.g. king lifted, king placed, rook lifted, rook placed for castling), not as a single jump to the final position.

Display

ManyaCynus's own screen shows the last move made (e.g. "E2-E4", "0-0" for castling), "POS OK" once the starting position is confirmed, and the specific offending square(s) (e.g. "+E4") when a scan doesn't

settle into a legal position.

Recovering from an unexpected reconnect

If ManyaCynus loses its BLE connection mid-game (e.g. it resets itself) and reconnects, the gateway would normally insist on seeing the exact starting position again before resuming, blocking the game in progress.

Manual override: lifting both Kings off the board at any point — right after reconnecting, or even mid-game — arms a manual override instead: the display shows "newpos", automatic rescanning stops, and the gateway waits (no time limit) for the desired position to be rebuilt. Once both Kings are back on the board, press ManyaCynus's own Clock/Scan button to submit that position as-is, no legality check — exactly what a real chess computer's own "set up position" mode expects.

4. Startup sequence

Follow this order every time, especially when using a Mephisto Phoenix or Chess Element module — both can take a few seconds to boot, and the gateway only gets one look at the cable during that window:

1. **Make sure the USB port has no power on it.** Don't have the gateway powered from USB (e.g. still plugged into a computer or power bank) at the same time it's powered from the chess computer module's cable — power the gateway from the module only, exactly as the wiring diagram in section 10 describes.
2. Power on the chess computer module and **wait until it has fully booted** before doing anything else.
3. **Only then** switch on the e-board (Chessnut, MILLENNIUM Supreme T2 BT, ManyaCynus, etc.).

If the module is unusually slow to boot and the gateway's own 5-second cable check elapses first, it falls back to standalone mode (section 5) — but this is self-correcting: as soon as the gateway sees any data arrive on the cable after that, even minutes later, it automatically drops any standalone BLE-to-BLE connection and restarts itself into normal cable mode. No manual power cycle is needed to recover from a slow-booting module.

5. Standalone mode (no cable computer required)

If no chess computer module is detected on the cable within 5 seconds of power-up, the gateway switches into a standalone mode: it keeps its normal BLE connection to the e-board, and **additionally** starts advertising itself as a BLE peripheral — masquerading as either a genuine ChessLink board or a genuine Chessnut board — so wireless chess software can connect directly to the gateway with no chess computer module in the loop at all. This is automatically undone (with a self-restart back into normal cable mode) the moment cable data is later detected, so it's safe even if a module just happened to boot slowly rather than being genuinely absent.

Setup sequence (all signalled on the connected e-board itself — LED squares light up on Millennium/Chessnut boards, ManyaCynus shows text):

1. The gateway connects to the e-board as usual and confirms its starting position.
2. **Ready signal:** the four center squares (d4/d5/e4/e5) light up (ManyaCynus: "OK") for 2 seconds, then clear.
3. Place a **second white queen** on the board (every one of these e-boards' piece sets includes a spare queen for promotion anyway):
 - **a4** selects ChessLink masquerade — the gateway advertises as "MILLENNIUM CHESS", the same name and protocol a real Millennium Supreme board uses.
 - **b4** selects Chessnut masquerade — the gateway advertises as "Chessnut Air", speaking Chessnut's own native BLE protocol.
4. **Confirmed signal:** the four corner squares light up (ManyaCynus: "CSLMode" / "NutMode") for 2 seconds, then clear, and the selected masquerade starts advertising.
5. Connect to the gateway from your chess software like you would to a real board of that type.

Masquerade	Confirmed working with
ChessLink ("MILLENNIUM CHESS")	PicoChess, BearChess, and at least one other independent ChessLink client
Chessnut ("Chessnut Air")	Chess PGN Master, BearChess

This mode is independent of, and doesn't change, normal cable operation — if a chess computer module is present on the cable at power-up, standalone mode never activates.

6. Supported chess computer modules (cable side)

Tested working: MILLENNIUM King and Mephisto Phoenix. Both speak Mode B over the cable, but not identically — the gateway auto-detects a module's exact checksum convention (plain 7-bit vs. odd-parity-encoded) from its first frame and matches it in its own replies, so no manual module selection is needed here either.

7. Automatic PGN recording

The gateway can automatically record every game played on the connected e-board as a PGN file. Recording works in every supported configuration:

- Cable-connected chess computer (Phoenix/King) with any e-board
- Standalone mode (ChessLink/Chessnut masquerade)
- Plain human-vs-human over-the-board play, with nothing else connected at all

The gateway runs its own self-contained chess rules engine (legal move generation, check/checkmate detection, standard SAN notation) purely from the sequence of board positions it observes — no external engine, app or GUI is required for recording itself.

A game is saved automatically when:

- **Checkmate** is detected — result recorded as `1-0` or `0-1`.
- The standard **starting position** is recognized again, meaning a new game has begun. The just-finished game is saved with result `*`, unless it was fewer than 10 full moves long, in which case it's discarded instead (almost always a setup/test, not a real game).

A manual result signal is also available mid-game: place both kings on one of three square pairs (a deliberate gesture, not a legal move):

Kings on	Result
d5 and e4	White wins (<code>1-0</code>)
d4 and e5	Black wins (<code>0-1</code>)
d4 and d5, or e4 and e5	Draw (<code>1/2-1/2</code>)

It doesn't matter which specific king sits on which square within a pair — only that the two kings together occupy exactly that pair. Manually-signaled and checkmate results are always saved regardless of game length.

The board itself also confirms the result for 3 seconds once the gesture is recognized: LED boards (Millennium, Chessnut) light the same d4/d5/e4/e5 squares steady (not blinking); ManyaCynus shows 1:0 , 0:1 , or 1/2:1/2 on its display.

The 20 most recently saved games are kept in a round-robin, oldest slot overwritten first once all 20 are in use — except a game is removed sooner, right after retrieval (section 8) has confirmed it was copied off the gateway successfully.

8. Retrieving saved games

Two independent ways to get saved games off the gateway:

Chess PGN Master, over BLE

Put the gateway into standalone Chessnut masquerade mode (section 5) and connect from Chess PGN Master like you would to a real Chessnut board — its own download feature retrieves saved games directly. Once a download completes, the gateway deletes that game from its own storage.

USB, via a standalone Windows tool

Trigger a **"queen gesture"**: set up the standard starting position with one extra white queen placed on c4 (every one of these e-boards' piece sets includes a spare queen for promotion anyway) — the corner squares light up to confirm it was recognized. Every saved game is then dumped over the gateway's native USB-CDC connection to a PC, where a small standalone Windows tool saves each one as its own `.pgn` file. Once the tool confirms every game saved correctly, the gateway deletes exactly those games from its own storage — a dump that never reaches the tool, or a partial transfer, leaves every game in place for the next attempt. See the `pgntool/` folder in the repository for the tool itself and full usage instructions.

Both retrieval paths are confirmed working — use whichever is more convenient.

9. Firmware updates (web installer)

Flash the gateway firmware directly from a supported browser (Chrome or Edge), no toolchain install required:

<https://parlue.github.io/BluetoothMax/>

Connect an ESP32-C3 SuperMini (or compatible board) via USB and follow the on-page instructions.

10. Hardware: components, wiring & safety

Components

- ESP32-C3 SuperMini
- HW-027 RS-232-to-TTL module with MAX3232
- 9 V to 3.2 V DC/DC step-down converter (Pololu D36V6F3)
- 4-pin Mini-DIN plug or suitable connection cable
- Wires and suitable plug or solder connections
- Optional enclosure, strain relief and insulation material

ChessLink cable side — pinout

Confirmed-correct pin assignment for the 4-pin Mini-DIN connector, viewed face-on into the socket, keyway/notch at the bottom, guide pin at the top:

Clock position	Wire color	Function
1 o'clock	Yellow	+9 V supply (from the chess computer module)
5 o'clock	Black	RS-232 IN (into the level-shifter's input)
7 o'clock	Green	GND
11 o'clock	Red	RS-232 OUT (from the level-shifter's output)

If you're building your own cable, trust this table over any datasheet-style "pin 1/2/3/4" numbering, and verify against a multimeter/oscilloscope before trusting a new build.

Serial settings

- 38400 baud
- 7 data bits
- Odd parity
- 1 stop bit (701)

Wiring diagram — power

Module +9 V	> DC/DC IN+
Module GND	> DC/DC IN-

```
DC/DC OUT 3.2 V ||> ESP32-C3 3V3
DC/DC GND      ||> ESP32-C3 GND

ESP32-C3 3V3    ||> HW-027 VCC (+), TTL side
ESP32-C3 GND    ||> HW-027 GND (-), TTL side
```

All components must share a common ground. The 9 V cable supply must never be connected directly to an ESP32 GPIO or to the ESP32 3.3 V pin.

Wiring diagram — data lines

```
Module TX (RS-232 OUT, see pinout table)
  ||> HW-027 RS-232 input
  ||> HW-027 TTL output
  ||> ESP32-C3 GPIO20 (RX)

ESP32-C3 GPIO21 (TX)
  ||> HW-027 TTL input
  ||> HW-027 RS-232 output
  ||> Module RX (RS-232 IN, see pinout table)
```

GPIO20 is used exclusively as the receive path from the chess computer module. GPIO21 is the transmit path to the module.

Important safety notes

- Connect the ESP32 only to the **TTL side** of the HW-027.
- Verify supply voltages and ground with a multimeter before connecting the hardware.
- Power the TTL side of the HW-027 with 3.3 V.
- Never connect RS-232 levels directly to an ESP32 GPIO.
- Check the viewing direction of Mini-DIN connectors: the solder side and plug side are mirrored, and the pinout table above is specified face-on into the socket — double-check orientation before trusting a measurement.
- Disconnect power before changing any wiring.

11. Version history

Version	Adds
v1.0	MILLENNIUM ChessLink board support (single-board)
v2.0	+ Chessnut Air/GO/Pro support (multi-board)
v3.0	+ Mephisto Phoenix chess computer support (cable-side checksum auto-detect)
v4.0	+ ManyaCynus robot support (castling, en passant, promotion)
v5.0	+ Standalone mode: BLE-to-BLE ChessLink/Chessnut masquerade, no cable chess computer required
v5.1	Fix: Chessnut+Mephisto Phoenix status checksum (single-square changes and captures could silently fail to register); standalone mode now recovers automatically if a cable module boots slowly instead of needing a power cycle
v5.2	Fix: BearChess's ChessLink move-suggestion LEDs (a ghost-square filter built for Mephisto Phoenix's own reset splash was wrongly eating small real move suggestions from BearChess)
v6.0	+ Automatic PGN game recording with USB retrieval (queen-gesture trigger, standalone Windows tool, delete-on-confirmed-transfer)
v6.1	Fix: the manual king-gesture result signal could be lost entirely if its target squares (d4/d5/e4/e5) were occupied by other pieces or the two kings didn't arrive together
v6.2	Chess PGN Master's BLE game download now actually works and properly deletes each game from the gateway after a successful pickup; + initial (real-hardware-untested) iChessOne board driver
v6.3 (current)	Manual king-gesture game results are now also confirmed on the board itself (3s LED/display signal); + ManyaCynus manual position override to recover from an unexpected mid-game BLE reconnect without losing the game

12. Contact, license & trademark notice

Contact

Board manufacturers who would like their board supported, and anyone who'd like a ready-built device but can't solder one themselves, are welcome to get in touch: **dsommerfeld@mac.com**

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